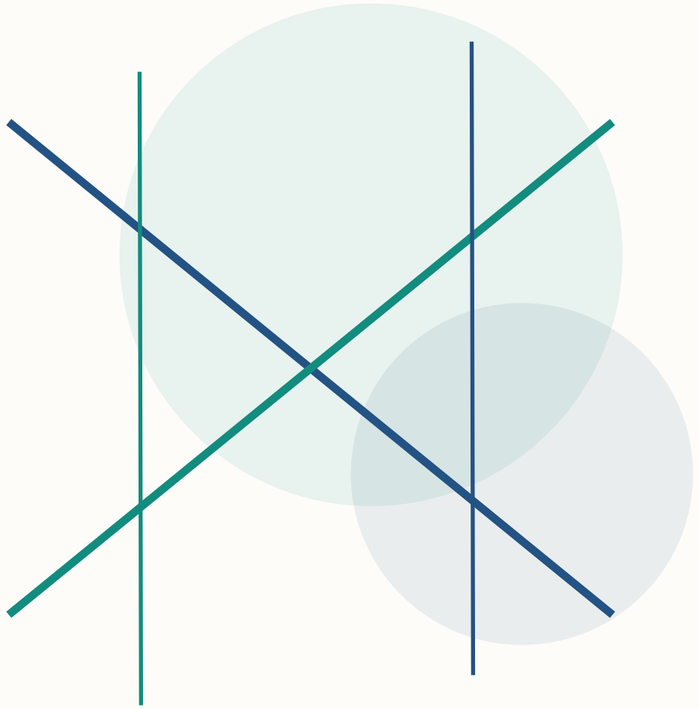
Build Your First Textile Web App



GitHub + Streamlit for Beginners

Build FiberX — a simple textile fiber explorer — and publish it as a live web app.

Idea → AI → Code → Files → GitHub → Streamlit → Live App

Today’s Goal

Leave with a working mental model — and a path to a live app.



By the end, you can explain:

- how an idea becomes a small app
- why three project files matter
- how GitHub and Streamlit work together

YOUR TURN: BUILD, NOT MEMORIZE

Meet FiberX

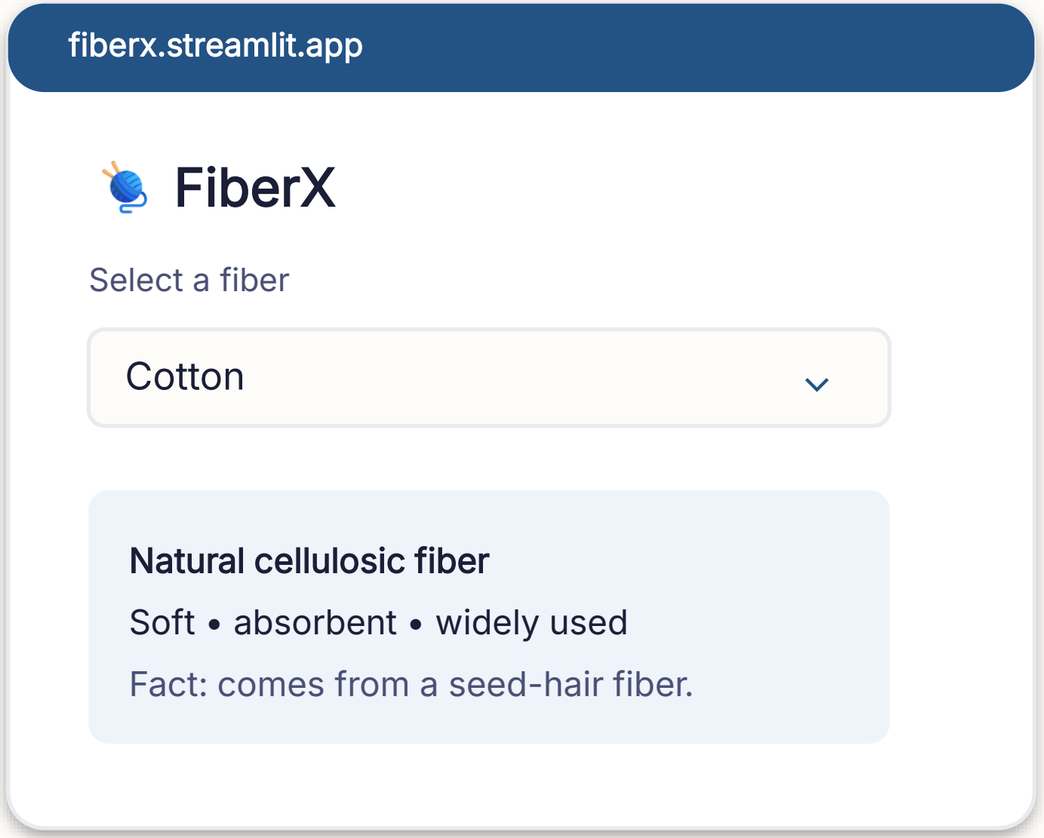
A simple explorer for textile fiber facts — built for learners, not programmers.

What FiberX shows

Classification · properties · advantages

Limitations · end uses · producing countries

A short, memorable fiber fact



Today’s Journey

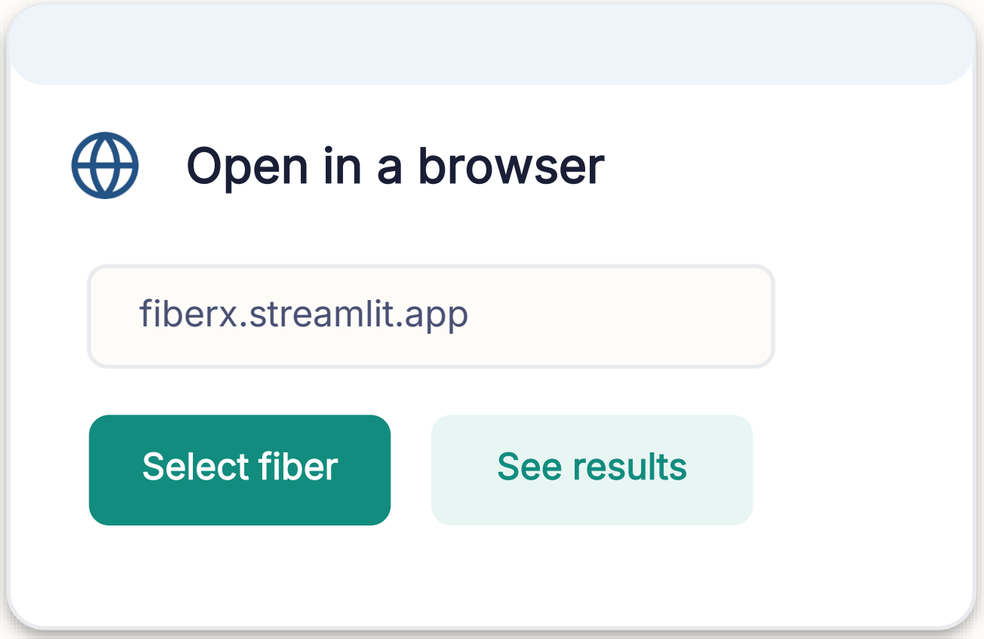
A map for the whole workshop — no mystery steps.



Think like an engineer: define → test → improve.

What Is a Web Application?

A useful tool that people **use in a web browser** — instead of installing it like a traditional desktop program.



FiberX is a web app: select a fiber, then see useful textile information immediately.

What Is Streamlit?

Streamlit lets us turn Python code into an interactive web application.

Write simple
Python instructions



Get interactive
browser widgets

Fast prototype
for FiberX

Source: Streamlit documentation — Python scripts can become interactive web apps and be deployed on Community Cloud.

What Can Streamlit Create?

Small building blocks that become an interactive research tool.

Inputs
select / type

Buttons
actions

Tables
compare facts

Charts
visualize data

Text
explain results

FiberX begins with one input: “Select a fiber.”

Why Streamlit for FiberX?

It gives a fast path from a small idea to something classmates can try.

Python-based
One readable language for logic + interface.

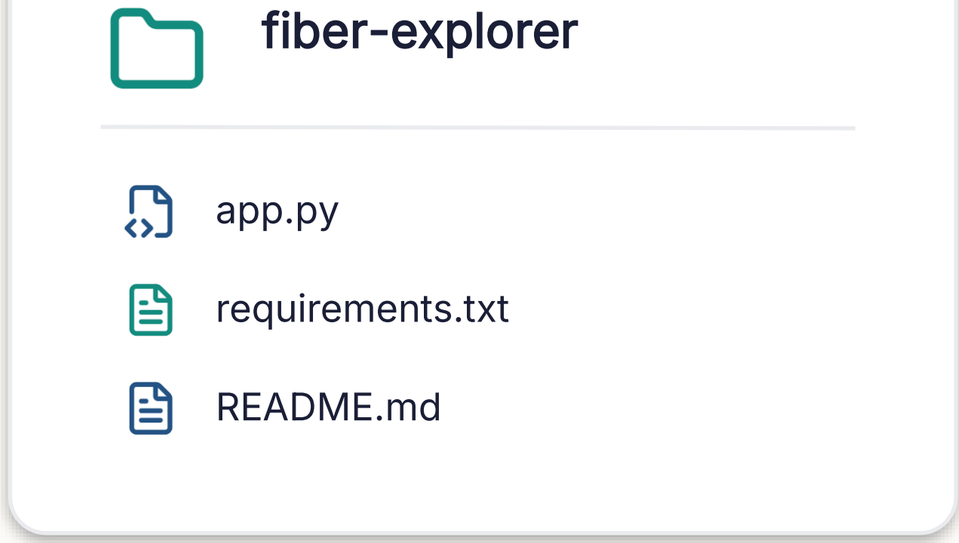
Fast
See an app without building a full website.

Beginner-friendly
Start with a few clear instructions.

Deployable
Share a live link through Community Cloud.

What Is GitHub?

GitHub stores and manages our **project files online.**



Think of it as the online home for FiberX — a place where your files stay together.

Source: GitHub Docs — repositories store projects; commits save snapshots of changes.

What Is a Repository?

Repository = project folder on GitHub.

fiber-explorer

Holds your files
Remembers changes
Can connect to Streamlit

One project → one organized home

A repository makes your app easy to find and share.

Why Use GitHub?

GitHub gives your first app an online home with a useful memory.



Storage
Files in one place



Changes
Snapshots of work



Collaboration
Share and improve



Deployment
Connect to Streamlit

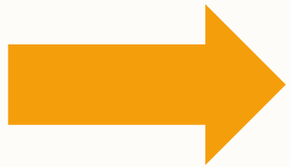
A “commit” is a saved snapshot of your project files.

GitHub vs Streamlit

They are teammates — not alternatives.

GitHub

Stores project files
Tracks your changes
Keeps the project organized



connects to

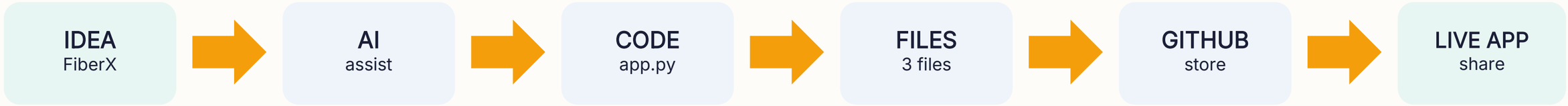
Streamlit

Runs your app
Creates a live web page
Gives you a shareable URL

GitHub stores FiberX. Streamlit brings FiberX to life.

The Complete Workflow

Your project moves through a small chain of clear hand-offs.



AI can: generate, explain, modify, debug.
Use it as a helpful collaborator during each step.

The engineer must: define, check, verify, decide.
AI can write code, but **YOU must check it.**

What Files Do We Need?

For our first app, just three files create a complete project.



app.py
The main application



requirements.txt
The needed packages



README.md
The project guide

Small project. Clear files. One powerful learning loop.

FiberX Project Structure

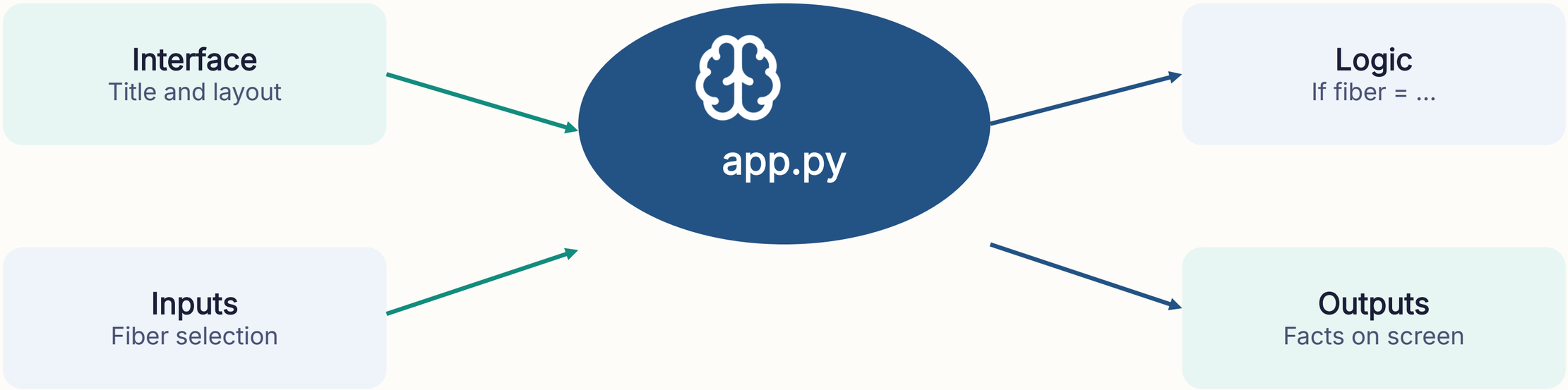
Keep every starter file in the same project folder.



Folder organization is an engineering skill — start it early.

app.py Is the Brain

This is the file where FiberX’s interface and decisions are described.



A Small, Readable app.py

```
import streamlit as st
st.title("🧵 FiberX")
fiber = st.selectbox(
    "Select a fiber",
    ["Cotton", "Polyester", "Wool"]
)
if fiber == "Cotton":
    st.write("Natural cellulosic fiber")
elif fiber == "Polyester":
    st.write("Synthetic fiber")
elif fiber == "Wool":
    st.write("Natural protein fiber")
```

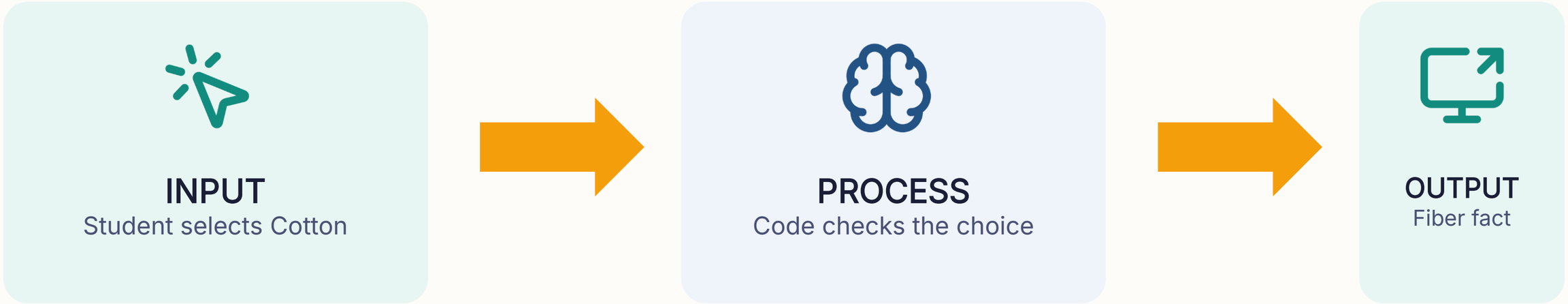
Readable first
Small code can teach big ideas.

Check it
Test each fiber selection.

You do not need to memorize every line today.

Input → Process → Output

Every interactive app can be understood with this simple pattern.



Selection → decision → useful textile information

requirements.txt

This small file tells the deployment system which Python packages the app needs.



“What do I need to run?”

`requirements.txt`

It lists the tools needed by FiberX.

Without it, Streamlit may not know what to install before running your app.

A One-Line requirements.txt

The first FiberX app depends on only one package.

requirements.txt

```
streamlit
```

Type it exactly. Lowercase. No extra explanation.

Later projects may need more packages. Start simple, then add only what your app truly uses.

README.md

This file explains what your project is and how someone can use it.



Project guide

What FiberX does
How to run it
Who it helps

```
# FiberX
A textile fiber explorer.
Select a fiber to see basic
classification and properties.
```

A good README helps people understand your work before they open your code.

Why README?

Because every project needs a human-friendly front door.



Future you
Remember the project



A classmate
Understand it quickly



A reviewer
See the purpose first

Good engineering is not only making it work — it is making it understandable.

BEGINNER RULE

Do we need more files?

Not for our first app.

Start with the smallest useful project.
Add files only when the app truly needs them.



Small start
→ steady growth

Avoid “tool overload.” Learn the core loop first, then expand with purpose.

Three Files, One App

Each file has one job. Together, they make FiberX usable and shareable.



Clear code + declared tools + clear instructions = solid starter project

Restaurant Analogy

A familiar way to remember what each part does.

app.py = kitchen
makes the meal

requirements = ingredients
tools you need

README = menu
explains the offering

GitHub = restaurant address
where it is stored

Streamlit = serving table
brings it to people

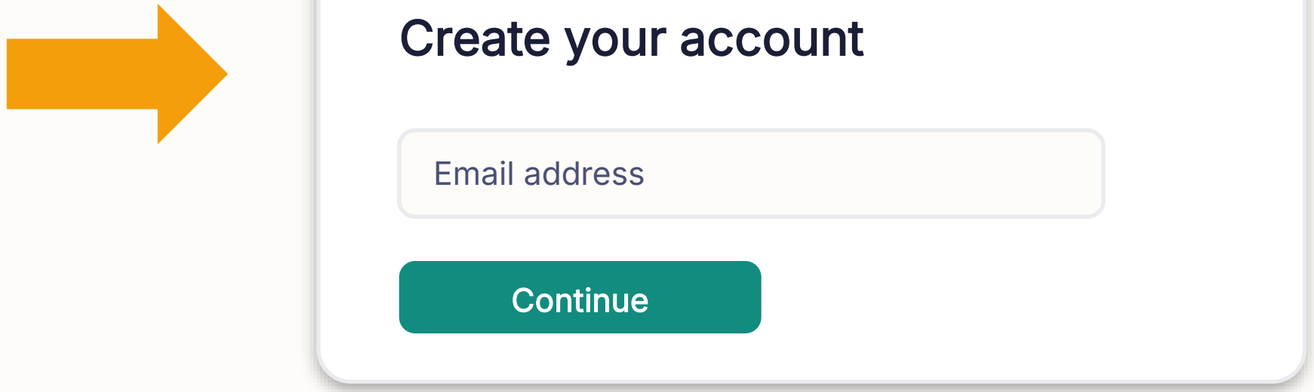
Different roles. One experience for the person using FiberX.

YOUR TURN

Step 1 — Create a GitHub Account

Your online home for the FiberX project.

1. **Open github.com**
2. Choose Sign up
3. Use an email you can access

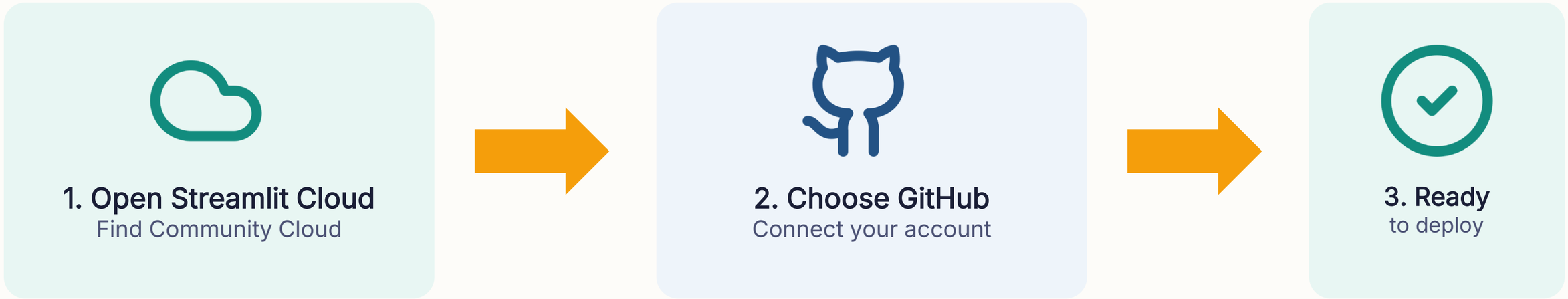


Tip: choose a simple username you would be comfortable sharing with classmates.

YOUR TURN

Step 2 — Create a Streamlit Account

Use Streamlit Community Cloud and sign in with GitHub.



Community Cloud connects directly to GitHub repositories and can deploy an app in minutes. Source: Streamlit documentation.

CHECKPOINT

Accounts ready?

Before we build, confirm both doors are open.

GitHub
Signed in successfully

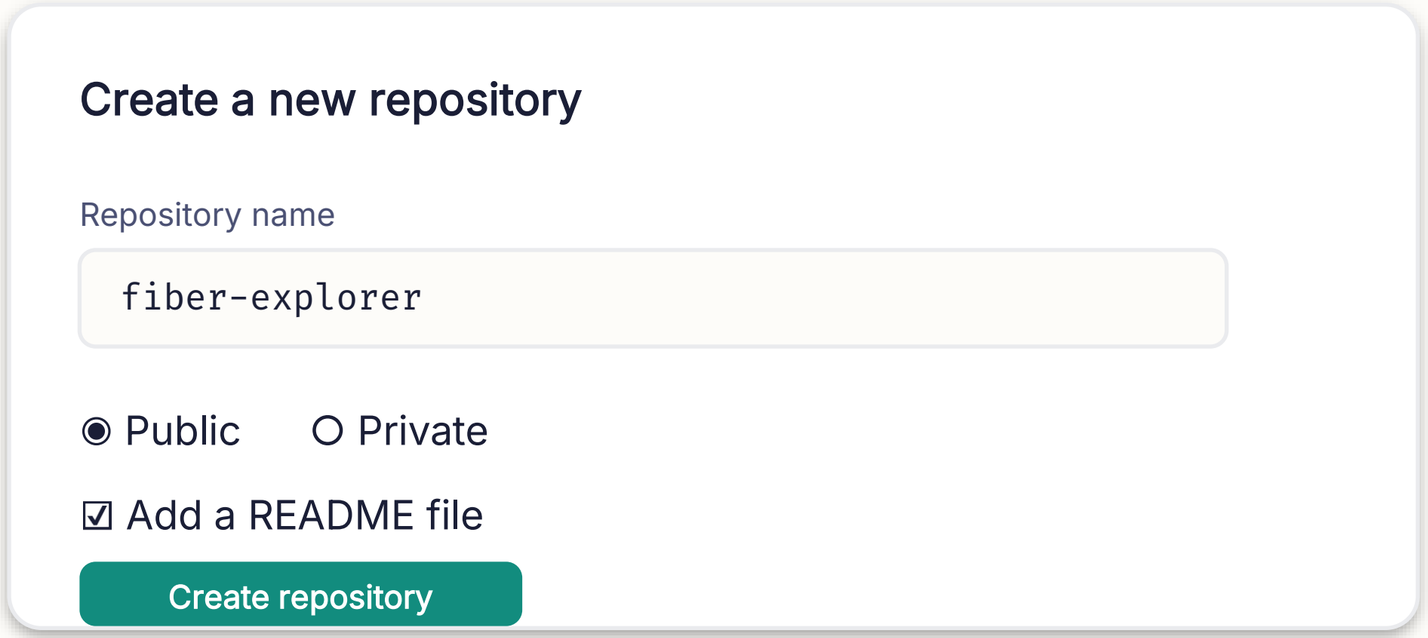
Streamlit Cloud
Connected to GitHub

If either is not ready, pause here — then rejoin the next step.

YOUR TURN

Create the fiber-explorer Repository

A repository is the container for every FiberX file.



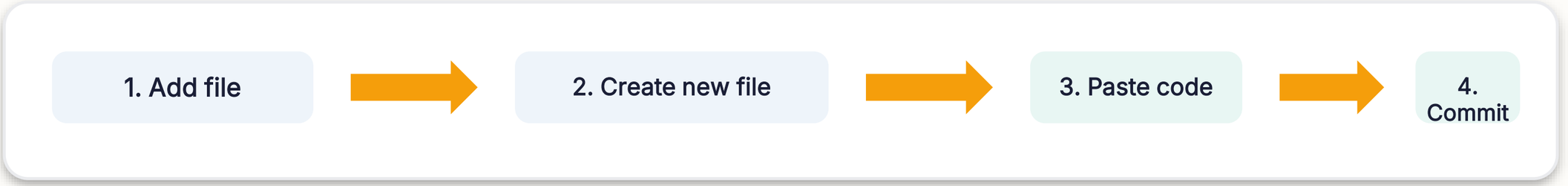
Name it clearly.
Use lowercase + hyphens.

GitHub Docs recommends a short, memorable repository name; README is a helpful starting point.

DEMO

Add app.py

Use GitHub’s web editor — no command line needed today.



File name: app.py
Commit message: Add FiberX app

When debugging with AI, provide:
1. Code · 2. Error message · 3. Expected result · 4. Actual result

YOUR TURN

Add requirements.txt

Make one more file in the same repository.

New file name

`requirements.txt`

File content

```
streamlit
```



Commit your file.

Now Streamlit will know what to install.

Exact spelling matters: requirements.txt is all lowercase and contains a dot before txt.

YOUR TURN

Add README.md

Tell a future reader what FiberX is before they inspect any code.

```
# FiberX

A beginner-friendly textile fiber explorer.
Select Cotton, Polyester, or Wool
to see classification and basic facts.
Built with Streamlit.
```



Keep it short.
Clear beats clever.

If GitHub created README.md when you made the repository, edit that file instead of making a duplicate.

CHECKPOINT

Repository complete?

Your GitHub repository should contain exactly these starter files.



fiber-explorer/



app.py



requirements.txt

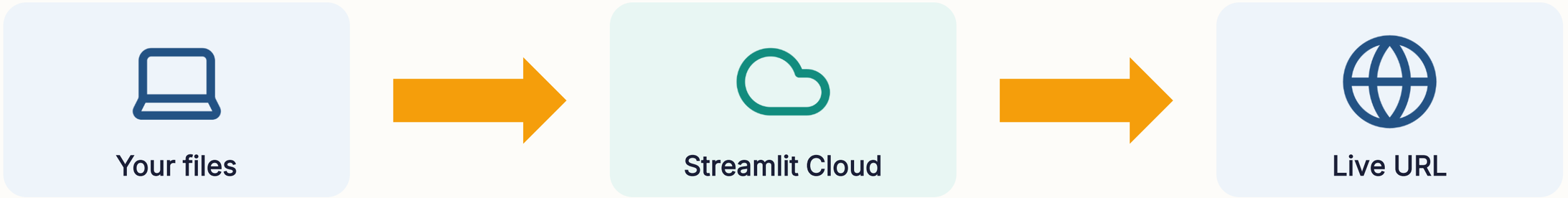


README.md

All three visible? You are ready to deploy.

Deployment: What Does It Mean?

Deployment means taking your project and making it available as a **live web app.**

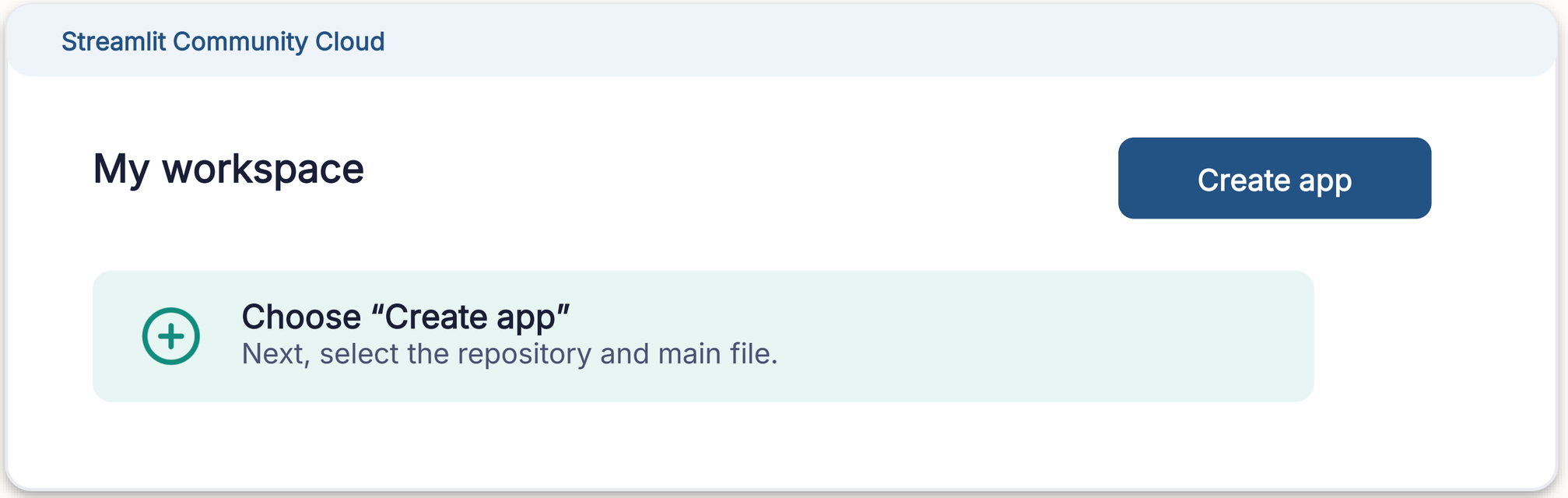


You are not sending an app to a distant planet; you are making it reachable in a browser.

DEMO

Streamlit: Create App

In Community Cloud, start a new app and point it at your GitHub repository.

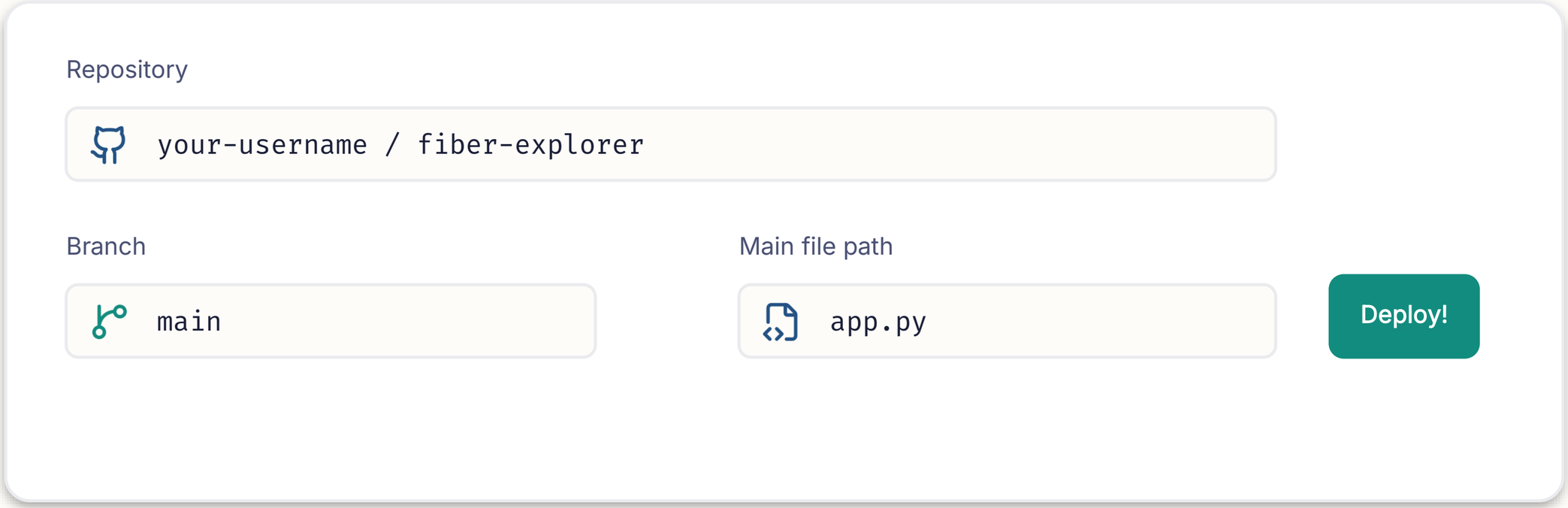


The exact screen may change, but the key choice remains: connect your repository to a new app.

DEMO

Choose Repository, Branch, and File

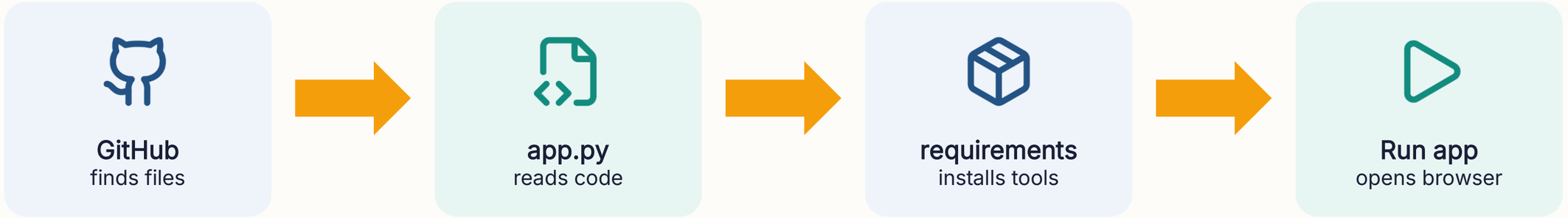
Tell Streamlit exactly where the FiberX instructions live.



Check these three fields carefully. A wrong repository, branch, or file path is a common beginner mistake.

What Happens During Deployment?

The cloud follows the instructions you prepared in GitHub.



Result: a live URL people can open and use.

If deployment fails, return to the three basics: file names, file contents, and the app error message.

DEMO

Your App Is Live

You now have a browser link for your textile engineering mini-project.

https://your-name-fiber-explorer.streamlit.app

FiberX
Textile Fiber Explorer

Select a fiber

Share the link.
A live app is easier to show, test, and improve.

Your live URL will be different — that is normal.

DEMO

Test FiberX

Try every option — then check both the code and the information.

Select and test

Cotton • Polyester • Wool

Does each choice show the right classification?
Does the output change when you select another fiber?

Verify the data

For production-country information, ask:
What is the source?
What year?
What exactly is being measured?

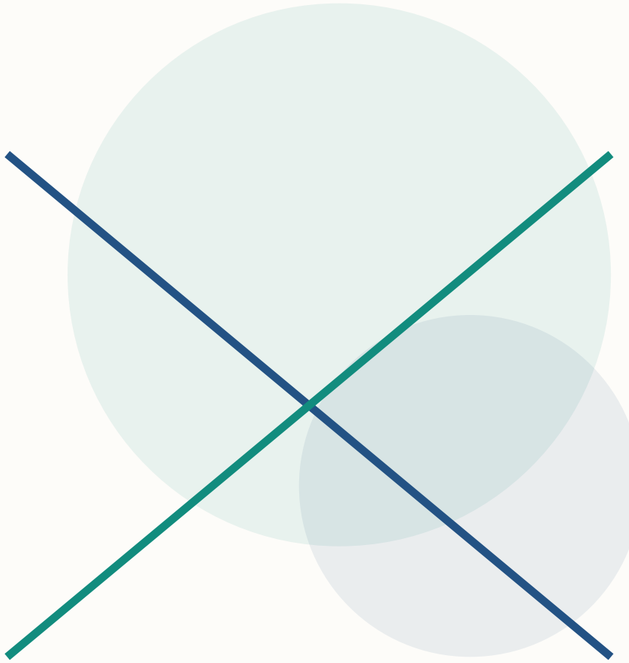
Cotton, polyester, and wool entries are learning examples. Before publishing specific country claims, verify a current primary or industry source.

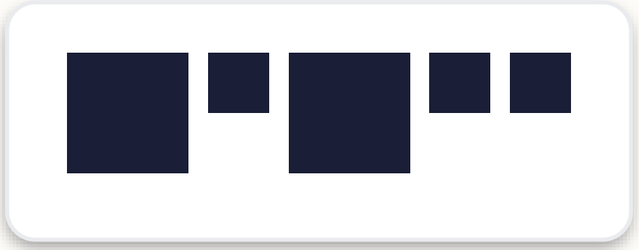
FINAL CHALLENGE

Build on FiberX.

Add **ONE** new textile fiber: Silk, Jute, Hemp, or Acrylic.

Use AI to draft code or explain a change.
Verify the fiber information before you publish it.





Live app link placeholder

AI is your collaborator, not your replacement.

Idea → AI → app.py → requirements.txt → README.md → GitHub → Streamlit → Live Web App